

# N6Cam — Test Report

**Date:** 2026-05-26 07:05:31  
**Device:** SIANA N6Cam (STM32N657 / IMX335 / Neural-ART NPU)  
**Firmware:** 01.08.2593089169  
**MCU UID:** 003031564236500400420024  
**Shell TTY:** /dev/ttyACM0  
**Runtime:** 42 seconds

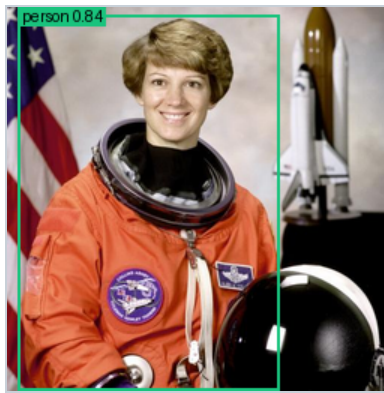
**Result:** ALL TESTS PASSED — Total: 56 | Pass: 56 | Fail: 0 | Skip: 0

| Test ID | Description                                      | Result |
|---------|--------------------------------------------------|--------|
| T00.1   | Kit CDC port /dev/ttyACM0 exists and is readable | PASS   |
| T00.2   | Shell `help` lists user-defined commands         | PASS   |
| T00.3   | `fw` returns FW VER: ...                         | PASS   |
| T00.4   | `uid` returns MCU UID                            | PASS   |
| T00.5   | `uptime` returns UPTIME ms                       | PASS   |
| T01.1   | `echo query` reports state + ok ack              | PASS   |
| T01.2   | `echo off` succeeds (ok-ack)                     | PASS   |
| T01.3   | After `echo off`, no echoed prompt before reply  | PASS   |
| T01.4   | `echo on` restores echo + ok ack                 | PASS   |
| T02.1   | `rtc` returns DD/MM/YY HH:MM:SS                  | PASS   |
| T02.2   | `rtc set DDMMYYYYHHMMSS` accepts a valid value   | PASS   |
| T02.3   | RTC reads back the value we just set             | PASS   |
| T02.4   | `version` prints Application:                    | PASS   |
| T03.1   | `irled on` reports state=1 + ok-ack              | PASS   |
| T03.2   | `irled off` reports state=0                      | PASS   |
| T03.3   | `motion sense 65 45` accepted, echoed back       | PASS   |
| T03.4   | `motion query` returns persisted 65/45           | PASS   |

|       |                                                                                 |      |
|-------|---------------------------------------------------------------------------------|------|
| T04.1 | `img quality 75` set + ok-ack                                                   | PASS |
| T04.2 | `img color RGB` accepted                                                        | PASS |
| T04.3 | `img chroma 1` accepted                                                         | PASS |
| T04.4 | `img query` shows quality=75 color=RGB chroma=1                                 | PASS |
| T05.1 | `detect start` succeeds                                                         | PASS |
| T05.2 | `detect profile 3 1` (people+vehicles, save-SD)                                 | PASS |
| T05.3 | `detect profile query` returns 0x03 / 0x01                                      | PASS |
| T06.1 | `notify enable 0x10` (people-detect bit) accepted                               | PASS |
| T06.2 | `notify period 30` accepted                                                     | PASS |
| T06.3 | `notify query` shows mask=0x10 period=30                                        | PASS |
| T06.4 | `notify trigger 16` emits +SDVRNTF JSON with rsn=16                             | PASS |
| T06.5 | `notify disable` succeeds                                                       | PASS |
| T07.1 | `sd query` reports mounted                                                      | PASS |
| T07.2 | `sd ls` returns at least one entry                                              | PASS |
| T07.3 | `photo savesd` returns SoW §7 filename<br>4325412_26052026_070458.rdy           | PASS |
| T07.4 | New .rdy file appears on SD after `photo savesd`<br>4325412_26052026_070458.rdy | PASS |
| T08.1 | `frame clear` reverts to live camera                                            | PASS |
| T08.2 | Upload a synthetic 192x192 grey frame                                           | PASS |
| T08.3 | `frame query` reports loaded                                                    | PASS |
| T08.4 | `frame run` returns a count + NN time on a grey frame<br>count=0 NN=88.2ms      | PASS |
| T09.1 | astronaut.jpg (real person) → NN detects ≥1 person<br>count=1 NN=88.3ms         | PASS |



input (256×256)



kit detections

camera.jpg (real person) → NN detects  $\geq 1$  person  
count=1 NN=88.3ms

T09.2



input (256×256)

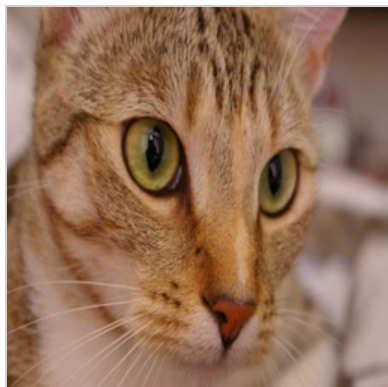


kit detections

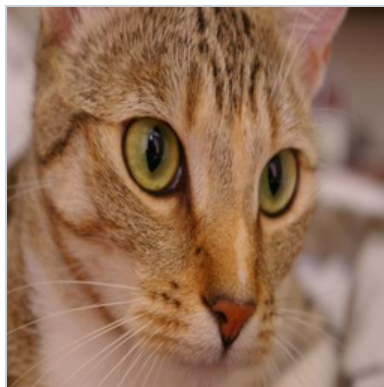
PASS

cat.jpg (no person) → NN returns 0 detections  
count=0 NN=88.3ms

T09.3



input (256×256)



kit detections

PASS

T09.4 coffee.jpg (no person) → NN returns 0 detections  
count=0 NN=88.3ms

PASS



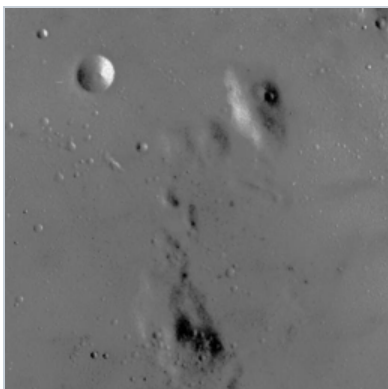
input (256×256)



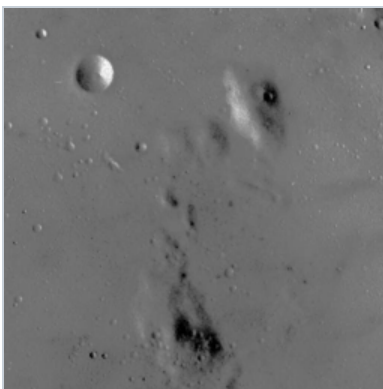
kit detections

moon.jpg (no person) → NN returns 0 detections  
count=0 NN=88.3ms

T09.5



input (256×256)

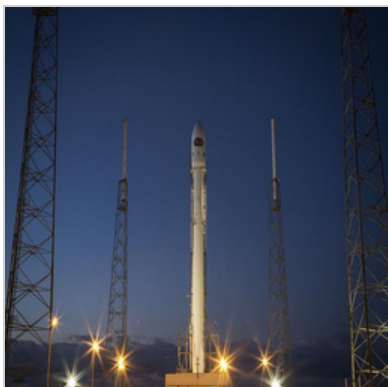


kit detections

PASS

rocket.jpg (no person) → NN returns 0 detections  
count=0 NN=88.3ms

T09.6



input (256×256)



kit detections

PASS

T09.7

NN inference avg ≤ 150 ms (SoW ≥ 5 FPS) over 6 images  
avg=88.3ms worst=88.3ms n=6

PASS

T10.1

`camera awb auto` accepted

PASS

T10.2

`camera exposure 10000` accepted

PASS

T10.3

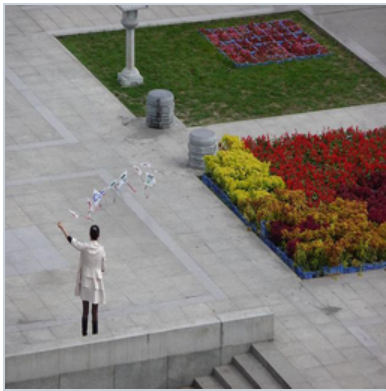
`camera gain 0` accepted

PASS

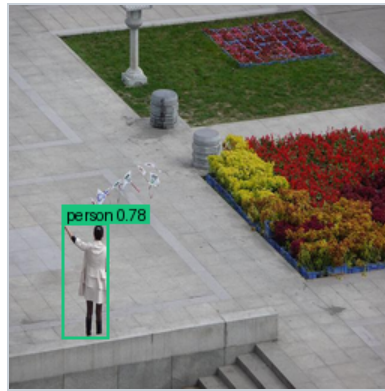
T11.1

1\_person.jpg (single person) → ≥ 1 person  
detected: person=1 vehicle=0 (gt 1p/0v) NN=88.4ms

PASS



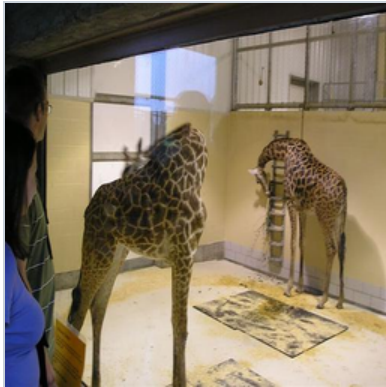
input (256×256)



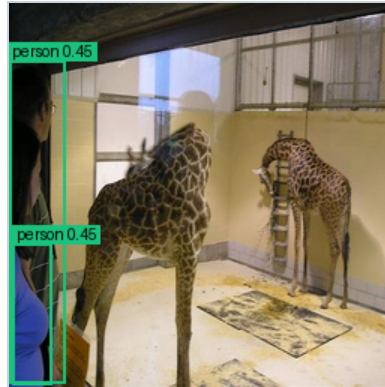
kit detections

2\_people.jpg (small group) → ≥1 person  
detected: person=2 vehicle=0 (gt 2p/0v) NN=88.4ms

T11.2



input (256×256)



kit detections

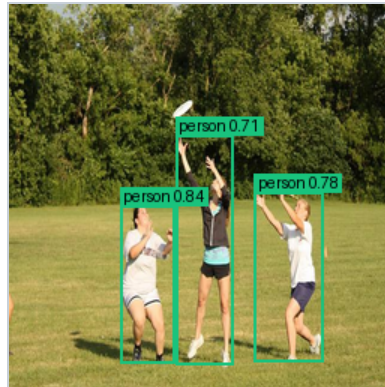
PASS

3\_people.jpg (small group) → ≥2 person  
detected: person=3 vehicle=0 (gt 3p/0v) NN=88.4ms

T11.3



input (256×256)



kit detections

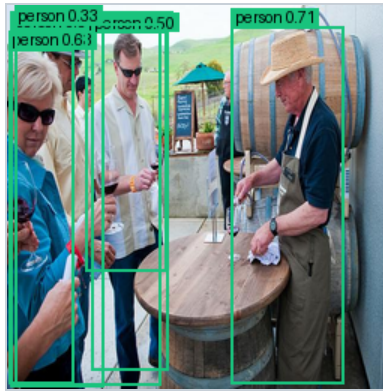
PASS

T11.4 7\_people.jpg (group) → ≥2 person  
detected: person=6 vehicle=0 (gt 7p/0v) NN=88.5ms

PASS



input (256×256)



kit detections

cars\_real.jpg (cars + people) → ≥1 person  
 detected: person=1 vehicle=3 (gt 3p/3v) NN=88.5ms

T11.5

PASS



input (256×256)



kit detections

cars\_real3.jpg (cars only) → ≥0 person  
 detected: person=0 vehicle=7 (gt 0p/6v) NN=88.5ms

T11.6

PASS



input (256×256)



kit detections

T11.7

13ppl\_5trucks.jpg (mixed cars+people) → ≥3 person  
 detected: person=7 vehicle=3 (gt 13p/8v) NN=88.5ms

PASS





input (256×256)



kit detections

cars\_15.jpg (vehicle: trains (not cars)) → ≥0 person  
detected: person=0 vehicle=1 (gt 0p/15v) NN=88.5ms

T11.8



input (256×256)



kit detections

PASS

cars\_18.jpg (vehicle: airplane (not cars)) → ≥0 person  
detected: person=0 vehicle=1 (gt 0p/18v) NN=88.5ms

T11.9



input (256×256)



kit detections

PASS